

DEPARTMENT OF COMPUTER APPLICATIONS
PSG COLLEGE OF TECHNOLOGY
MCA G1 & G2, SEMESTER 1
26MX16 - PROGRAMMING AND DATA STRUCTURES LABORATORY
PRACTICE SHEET 5

Prerequisite	2D traversal (Integer and String), Row traversal, Column traversal, Nested traversal, searching, position tracking, indexing and matrix transformation
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1. Data Center Server Load Matrix

A data center records the CPU usage of R servers across C time intervals. Store the values in a 2D array and calculate the total CPU usage of each server.

Input Format

- First line: R C
- Next R lines: C integer CPU values

Sample Input

```
3 4
40 50 60 70
80 75 65 70
30 45 50 55
```

Sample Output

```
Server 1: 220
Server 2: 290
Server 3: 180
```

2. IT Helpdesk Ticket Matrix

An IT organization stores the number of support tickets received by different departments over several days. Find the department with the highest total number of tickets. **Sample**

Input

```
3 5
10 15 12 20 18
25 20 30 15 10
5 8 10 12 15
```

Sample Output

```
Department 2
Total Tickets: 100
```

3. Server Failure Locator

A monitoring system stores server status using:

1 → Working

0 → Failed

Find the number of failed servers in each time interval.

Sample Input

```
4 5
1 1 0 1 0
1 0 0 1 1
1 1 1 1 0
0 1 0 1 1
```

Sample Output

```
Time 1: 1 failure
Time 2: 2 failures
Time 3: 3 failures
Time 4: 0 failures
```

Time 5: 2 failures

4. Search an Employee ID

An organization stores employee IDs in a 2D array representing different teams. Given an employee ID, find its team number and position.

Sample Input

3 4

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101 102 103 104

201 202 203 204

301 302 303 304

203

Sample Output

Employee Found

Team: 2

Position: 3

5. Rotate a Matrix

A cloud application stores server allocation data in a square matrix. Rotate the matrix 90 degrees clockwise.

Sample Input

3

1 2 3

4 5 6

7 8 9

Sample Output

7 4 1

8 5 2

9 6 3

6. Find the Highest-Load Server

A cloud platform stores CPU utilization of servers in a matrix.

Find:

- Highest CPU value
- Server number
- Time interval

Sample Input

3 4

45 60 55 70

80 75 65 90

50 55 85 60

Sample Output

Highest CPU Usage: 90

Server: 2

Time Interval: 4

7. Product Inventory Matrix

An e-commerce company stores product quantities in a matrix where each row represents a warehouse and each column represents a product. Find the total quantity of each product across all warehouses.

Sample Input

3 4

20 30 40 50

15 25 35 45

10 20 30 40

Sample Output

Product 1: 45

Product 2: 75

Product 3: 105

Product 4: 135

8. Application Permission Matrix

An enterprise application maintains a permission matrix:

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1 → Access Granted

0 → Access Denied

Rows represent employees and columns represent applications.

Given an employee number, count the number of applications to which the employee has access.

Sample Input

4 5

1 1 0 1 0

1 0 1 1 1

0 0 1 0 1

1 1 1 1 1

2

Sample Output

Applications Accessible: 4

9. Word Search in an Employee Directory ★

An organization stores employee names in a 2D string array. Given a search string, determine whether the name exists. The search must be case-sensitive.

Sample Input

3 3

Arun Bala Charan

Divya Ezhil Farah

Gokul Hari Isha

Farah

Sample Output

FOUND

Row: 2

Column: 3

Sample Input 2

3 3

Arun Bala Charan

Divya Ezhil Farah

Gokul Hari Isha

Rahul

Sample Output 2

NOT FOUND

10. Find Duplicate Usernames ★

A software application stores usernames in a 2D array representing different servers. Find whether any username appears more than once.

Sample Input

3 3

admin user1 user2

user3 admin user4

user5 user6 user2

Sample Output

Duplicate Usernames:

admin

user2

11. Diagonal Server Analysis ★★

A monitoring system stores server performance scores in a square matrix. Calculate:

- Sum of the main diagonal

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- Sum of the secondary diagonal
- Difference between the two sums

Sample Input

4

10 20 30 40

50 60 70 80

90 15 25 35

45 55 65 75

Sample Output

Main Diagonal Sum: 170

Secondary Diagonal Sum: 160

Difference: 10

12. IT Department Performance Analyzer ★★

An organization records employee performance scores using a 2D array. Each row represents a department and each column represents an employee.

For every department:

- Find the highest score.
- Find the lowest score.
- Calculate the number of employees scoring above 80.

Sample Input

3 4

75 85 90 65

88 72 95 91

60 70 82 85

Sample Output

Department 1

Highest: 90

Lowest: 65

Above 80: 2

Department 2

Highest: 95

Lowest: 72

Above 80: 3

Department 3

Highest: 85
Lowest: 60
Above 80: 2

13. Smart IT Service Dashboard ★★☆☆

An IT service platform maintains a matrix containing the number of incidents handled by different support teams across several days. A high-load day is a day where the total number of incidents is greater than 100.

Determine:

- Total incidents handled by each team.
- The day with the highest number of incidents.
- Number of high-load days.
- The team with the highest total workload.

Sample Input

```
3 5
20 25 30 35 40
15 20 25 30 35
```

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```
30 25 20 35 40
```

Sample Output

Team 1 Total: 150

Team 2 Total: 125

Team 3 Total: 150

Peak Day: 5

Peak Day Incidents: 115

High-Load Days: 2

Highest Workload Team: 1

Skills Acquired	2D Array Traversal, Row/Column Processing, Nested Iteration, Integer and String Searching, Index and Position Tracking and Matrix Transformation.
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